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Assessment of online public opinions on large infrastructure projects: A case study of the Three Gorges Project in China

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ABSTRACT

Public opinion becomes increasingly salient in the ex post evaluation stage of large infrastructure projects which have significant impacts to the environment and the society. However, traditional survey methods are inefficient in collection and assessment of the public opinion due to its large quantity and diversity. Recently, Social media platforms provide a rich data source for monitoring and assessing the public opinion on controversial infrastructure projects. This paper proposes an assessment framework to transform unstructured online public opinions on large infrastructure projects into sentimental and topical indicators for enhancing practices of ex post evaluation and public participation. The framework uses web crawlers to collect online comments related to a large infrastructure project and employs two natural language processing technologies, including sentiment analysis and topic modeling, with spatio-temporal analysis, to transform these comments into indicators for assessing online public opinion on the project. Based on the framework, we investigate the online public opinion of the Three Gorges Project on China's largest microblogging site, namely, Weibo. Assessment results present spatial-temporal distributions of post intensity and sentiment polarity, reveals major topics with different sentiments and summarizes managerial implications, for ex post evaluation of the world's largest hydropower project. The proposed assessment framework is expected to be widely applied as a methodological strategy to assess public opinion in the ex post evaluation stage of large infrastructure projects.

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1. Introduction

The public opinion is crucial for decision making of large infrastructure projects and the public satisfaction is also a key indicator for ex post evaluation of large infrastructure projects (Chan and Chan, 2004; Ji and Hong, 2016; Stoutenborough and Vedlitz, 2014). Successful implementation of large infrastructure projects depends not only on the financial and environmental feasibility, but mainly on the support from the public, who ultimately pays for and will be influenced by these projects (Friedler et al., 2006). Thus, the assessment of public opinion should be included in the delivery framework of large infrastructure projects (Ng et al., 2013). However, the complexity of large infrastructure projects and diverse interests of the public stakeholders make the assessment of public opinion very difficult.

Large infrastructure projects require more investment, consume more public resources and have more profound impacts on local even national economy, society and environment than general projects (Zeng et al., 2015). Hence, they usually have a large number of public stakeholders with a wide range and complicated levels. According to the impacts from projects, public stakeholders can be divided into two categories, including local community and general public. Local communities are directly influenced by large infrastructure projects. On the one hand, local people can benefit from large infrastructure projects, as these projects provide many jobs and public services for them. But on the other hand, if any environmental pollution or safety accident occurs, local people would be the direct victims (Zeng et al., 2015). With these impacts, local people have the motivation to organize substantial opposition activities against the construction or operation of large infrastructure projects if their interests are harmed (Liu et al., 2016). As a result, the acceptance of local people is important to the success of a project. Interests and views of local people on large infrastructure projects are included in current project delivery framework and are fully discussed in the literature (Frynas, 2005; Jobert et al., 2007; Jones and Eiser, 2010; Musall and Kuik, 2011; Wüstenhagen et al., 2007).

The general public is not directly or just slightly influenced by large infrastructure projects, but they are also concerned with the economic and environmental performance of these projects. Comparing with local people, views and comments from the general public would be more objective and unbiased (Zeng et al., 2015). Hence, the public opinion on a large infrastructure project can reflect a more holistic evaluation on that project. In addition, discussion of large infrastructure projects is an important activity in the political life of the public (O'Faircheallaigh, 2010). With the progress on democracy, understanding the public opinion will be more and more critical to policy making in

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relation to large infrastructure projects. However, in previous practices, public opinions were often overlooked in the delivery framework of large infrastructure projects (Akintoye et al., 2003; Ng et al., 2013). A major reason is that public opinions are difficult to collect and analyze (Ji and Hong, 2016).

Social scientists usually use questionnaire surveys to measure public opinions on conflictive large infrastructure projects (Cihar and Stankova, 2006; Hu et al., 2014). However, traditional survey methods are known as costly, time-consuming and invasive (Kirilenko and Stepchenkova, 2014; Kirilenko et al., 2015). The booming popularity of social media platforms, such as Facebook, Twitter and Weibo (the biggest microblogging site in China), highlights a new opportunity to collect and assess public opinions. User-generated contents in these social media platforms provide massive amounts of data on various topics of social importance. These data make social media as a valuable resource for assessing public opinions on large infrastructure projects, which are controversial and have significant impacts to the environment and society.

This study proposes an assessment framework, which combines two natural language processing technologies, including sentiment analysis and topic modeling, to transform unstructured social media comments on large infrastructure projects into sentimental and topical indicators for enhancing practices of ex post evaluation and public participation. Recent studies regard the interactions among social media users as a large and distributed network of sensors that react to external events by reporting individual comments (Kirilenko and Stepchenkova, 2014; Kirilenko et al., 2015). Using social media content to investigate the public opinion is a non-invasive research method avoiding the complications of interactions with human subjects and has a simple data collection process and a high sampling rate (near 100%). With the massive social media contents, many scholars implement comprehensive analysis of public opinions on natural and social phenomena (Ashley and Tuten, 2015; Gainsbury et al., 2016; Kirilenko and Stepchenkova, 2014; Kirilenko et al., 2015; Quintelier and Theocharis, 2013; Sakaki et al., 2013; van de Belt et al., 2015; Williams, 2013). These applications highlight the potential of using social media data to investigate public opinion on controversial issues, including the construction or operation of large infrastructure projects.

The utility of social media data for public opinion research is augmented by recent data mining technologies, especially sentiment analysis (Pang and Lee, 2008) and topic modeling (Blei et al., 2003). Public comments on social media sites are mainly textual data, which are high-dimensional. Sentiment analysis can map these textual data into an emotional dimension. In other words, whether a comment expresses a negative or positive emotion can be determined, providing an indicator to measure public satisfaction. Topic modeling can uncover the intellectual structure of a large collection of documents. Hence, major topics of the public opinion on a certain issue can be extracted by topic modeling. In the ex post evaluation stage of a large infrastructure project, the government and the management team are also exactly interested in the overall distribution of public sentiment (positive or negative) and the major topics of the public opinion on that project.

2. Literature review

2.1. Ex post evaluation of large infrastructure projects

By definition, ex post means after the completion, as opposed to ex ante, meaning beforehand (Olsson et al., 2010). Ex post evaluation is a highly valuable tool for determining not only how successful a large infrastructure project may have been after the completion, but rather the long-term performance of the outcomes for the economy, society and the environment (Geurs and van Wee, 2006). Different from final project evaluations, which are performed at the time of a project's completion to assess whether or not it has achieved its intended goals and is compliant with contract terms, an ex post evaluation is conducted in the years after a project's official end date to assess relatively long-term effects (Regan et al., 2011).

Ex post project evaluation is a systematic assessment framework and can be conducted from different perspectives (Olsson et al., 2010). In practice, the socio-economic evaluation is the most common content of ex post project evaluation and has been reported in many previous studies (Hidalgo et al., 2013; Pilavachi et al., 2008; van den Bergh et al., 2007). The original socio-economic evaluation only focuses on direct benefits for users and costs of the project, which can be easily calculated as economic values (Dmytrenko, 1997). However, with the prevalence of the concept of sustainability in recent project development (Ugwu et al., 2006; Arce and Gullón, 2000), economic evaluation based on narrowly scoped cost-benefit analysis is now insufficient for ex post evaluation of large infrastructure projects (Nakamura, 2000).

Recent studies focus on how to measure indirect or non-economic effects of large infrastructure projects within the socio-economic evaluation framework. For instance, Banister and Thurstain-Goodwin (2011) use a three-level framework to quantify the non-transport effects of railway projects on economic growth, agglomeration and property values. Based on externality theory (Dahlman, 1979), Zheng et al. (in press) develops an ex post evaluation model which recognizes all positive externalities, such as flood control, grid benefits and tourist attraction, and negative externalities, such as intergenerational land loss and migrants' invisible losses, in a hydropower project's life cycle and convert these externalities into economic values. Nevertheless, the above-mentioned non-economic effects, such as environmental effects, economic promotion, and additional public services, are all objective factors. Subjective factors, including participants' satisfaction, user satisfaction and public satisfaction are still difficult to quantitatively measure within the socio-economic evaluation framework, but are also important in ex post evaluation of large infrastructure projects (Chan and Chan, 2004; Ji and Hong, 2016). These subjective factors are not only individual evaluation indicators, but also important references for determining the relative weights of objective factors in the evaluation framework (Ji and Hong, 2016).

To analyze the subjective factors, questionnaire surveys, interviews and other participatory approaches are often used (Becker et al., 2003; Chanchitpricha and Bond, 2013; Diez, 2001). Participants' satisfaction and user satisfaction are relatively easy to investigate with questionnaire surveys or interviews, as the number of participants (client, design team, construction team and supervision team) in the construction process and the number of end users of a project are relatively small (Parfitt and Sanvido, 1993; Torbica and Stroh, 2001). However, using traditional methods to measure the public satisfaction is relatively difficult, as both questionnaire surveys and interviews are costly, time-consuming and invasive in collecting data from a large sample of the public. With increasing public awareness on environmental protection (Feng and Reisner, 2011; Hu, 2012) and progressive democratic deliberation in public policy making (Lonsbury, 2014), large infrastructure projects will attract more attention from the general public and meanwhile the public satisfaction will be more important in ex post evaluation of large infrastructure projects (Cohen et al., 2014; Jiang et al., 2016a). This calls for new approaches to effective analysis of public satisfaction and opinions on large infrastructure projects.

2.2. Public participation approaches

Public participation is “the involvement of individuals and groups that are positively or negatively affected by a proposed intervention (e.g., a project, a program, a plan, a policy) subject to a decision-making process or are interested in it” (André et al., 2006, p.1). O'Faircheallaigh (2010) summarizes three major aims of public participation. First, public participation is an aid to decision making which is separate from the general public. Second, public participation ensures that the public can be joint decision-makers. Third, public participation reconstitutes the decision making structures. In a democracy, public participation is an

important mechanism to foster justice, equity and collaboration and an effective tool to gather data and information from the public about their natural and social environment, as well as about the relations they have with their environment (André et al., 2006). As a result, public participation provides a better analysis of proposals, leading to more creative development, more sustainable intervention and greater public satisfaction of the project (Tang et al., 2008).

Public participation can take different forms, from passive participation such as public notice, to consultative participation such as questionnaire surveys, public hearings and open-houses, to interactive participation such as workshop, negotiation, mediation and co-management (Higgs et al., 2008; Rowe and Frewer, 2004). Different levels of public participation can occur in different phases in the project delivery process, from initial analysis of the local community and notice of the proposed intervention, to project approval, construction and ex post evaluations (André et al., 2006). Although the benefits and forms of public participation have been clearly identified in both theoretical and practical terms, the implementation of specific public participation programs remains difficult (Hartley and Wood, 2005; Stewart and Sinclair, 2007). Both practitioners and researchers call for greater public participation approaches (O'Faircheallaigh, 2010).

Recent studies highlight the potential of using information technologies (IT) in promoting public participation (Conroy and Evans-Cowley, 2006; Cox et al., 2015; Higgs et al., 2008; Tang and Liu, 2016). The prevalence of Geographic Information System (GIS) and Global Positioning System (GPS) functions in electronic devices empowers citizens to review or report nearby environmental information. The geo-referenced information can be quickly shared by the prosperous social media services with a large number of users and subsequently receive corresponding comments also with geographic tags. IT-based participatory approaches provide a platform to collect grass-roots opinions by the crowdsourcing technology (Brown and Donovan, 2014) and this is particularly important to public participation of large infrastructure projects, as large infrastructure projects always attract widespread attention from the general public. Only with IT-based participatory approaches, can decision-makers receive public comments from the widest population and, analyze and respond to these comments in real time. However, the huge number of public comments raises new questions relating to the approaches of automated analysis, as it is impossible to manually review or reply to all these comments.

2.3. Online public opinion analysis

In the near future, public participation will take place primarily via the Internet and social media sites will be the major platform for online public participation (Bertot et al., 2012; Bonsón et al., 2012; Lee and Kwak, 2012). With regard to controversial public issues such as large infrastructure projects, already in recent days, massive data of public opinions are published on social media every day. These social media data are highly valuable in public opinion research and have been widely used as sources for monitoring public responses to natural and social phenomena. For instance, Williams (2013) reported 575 peer-reviewed publications in politics, geography and natural disaster management using social media data.

The nature of social media data is significantly different from traditional survey data. First, social media content is published continuously and often simultaneously with the event being discussed, while survey responses often reflect past experiences of respondents and are collected at discrete moments. Second, the sampled units are different. Survey data are collected from individuals, but the social media data are composited by posts. Third, the sample sizes are different. The sample size of social media data is usually much larger than that of survey data. Forth, the relevance to the research topic of the survey data or social media data is different. Survey data are directly relevant to the research topic as the questionnaire and the interview guide are always predefined causing that the survey data are well-structured. However, social media data are not necessarily relevant to the research topic, as social media data are usually collected by keyword search or tag search leading to unstructured data with open-ended topics (Schober et al., 2016). These characteristics cause that most studies based on social media data are exploratory research or with simple correlation analysis (Ashley and Tuten, 2015; Gainsbury et al., 2016; Gu et al., 2016; van de Belt et al., 2015).

The analysis of social media data can be performed from three dimensions. First is the post intensity, which measures how many messages related to a specific issue have been published in a certain time period and/or within a certain region. Previous studies demonstrate that the post intensity on social media directly reflects the extent of prevalence of the issue. For instance, Kirilenko and Stepchenkova (2014) investigate the spatial-temporal patterns of tweeting on climate change topic. Kirilenko et al. (2015) check the relationship between number of tweets on climate change and local weather and extreme events. He et al. (2013) use the post intensity data from Facebook and twitter to perform competitive analysis for the pizza industry.

The second dimension is the sentiment of the public opinion. For any post collected from social media, sentiment analysis methods can determine whether the sentiment of the post is positive, negative or neutral (Pang and Lee, 2008) and subsequently obtain the sentiment distribution of all the posts collected. The sentiment distribution can reflect the extent of public support or satisfaction on the concerned issue and has been discussed in previous studies. For example, Ceron et al. (2015) employ sentiment analysis on twitter data to monitor electoral campaigns in the United States and Italy. Nguyen et al. (2015) use sentiment trends on twitter to predict stock movement. Cao et al. (2014) develop a traffic sentiment analysis system to monitor public satisfaction on the transportation sector.

The final dimension is the topics within the public opinion. A publicly controversial issue, such as a large infrastructure project, can positively and negatively influence the environment and society from different aspects, which are reflected in the narratives of online comments. Hence, it is necessary to identify specific topics in the massive online comments. Topic modeling methods can derive latent topics in a large corpus (Blei et al., 2003) and have been applied in previous studies to analyze the public opinion. For example, Tvinnereim and Fløttum (2015) use structural topic modeling to investigate major concerns of Norwegians with climate change. Qian et al. (2016) propose a multi-modal event topic model to analyze the public opinion and to track major events on social media. The integration of topic modeling and sentiment analysis can help decision-makers find out the determinants of public satisfaction or dissatisfaction with the concerned issue.

3. Methods

Based on the literature review, the assessment of online public opinion on a large infrastructure project should be performed from three dimensions, including post intensity reflecting prevalence of discussion on the project, sentiment distribution reflecting public satisfaction with the project and topics revealing the detailed concerns of the public on the project. Post intensity can be simply calculated by basic statistics. Sentiment distribution is generated by sentiment analysis methods and topics are derived by topic modeling approach.

The following assessment framework, depicted in Fig. 1, provides detailed procedures, including data collection and preprocessing, sentiment analyzing, topic modeling and indicator building. Analysis processes were mainly implemented in R (Ihaka and Gentleman, 1996). The Three Gorge Project (TGP) was chosen as case study to fully describe the proposed method, as the TGP has been widely regarded as one of the most controversial infrastructure projects in China, even in the world.

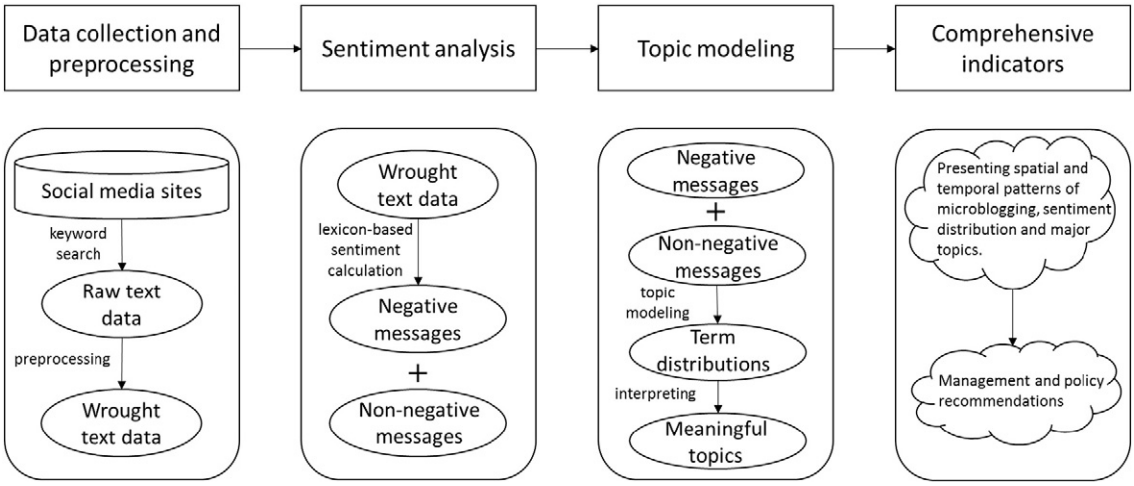


Fig. 1. Online public opinion assessment framework for the TGP.

3.1. Data collection and preprocessing

The TGP related Weibo messages were collected from Weibo.com, which is the biggest microblogging site in China and is known as China's twitter. Over 500 million users generate 100 million messages every day in Weibo. Weibo provides an official Application Program Interface (API) for keyword search. The API can print all Weibo messages which contain a given keyword and was published in a given time period. In June 2015, we used this API to collect all Weibo messages containing "三峡工程 (Three Gorges Project)" and/or "三峡大坝 (Three Gorges Dam, TGD)" and published in the time period from June 1, 2013 to May 31, 2015. In addition to message texts, meta information, including timestamp and location, were also collected. Finally, we got 171,615 Weibo messages satisfying the search criteria.

As Chinese texts do not segment words by spaces in sentences, segmentation is necessary for Chinese text mining. We employed a widely-used segmentation tool, the Institute of Computing Technology Chinese Lexical Analysis System (ICTCLAS) (Zhang et al., 2003) available in R, to segment words of Weibo messages. As the corpus contains many technical terms of hydropower, construction and environment, and many geographic names, we imported professional dictionaries downloaded from Sougou.com into the program to improve segmentation accuracy and effectiveness. Some Weibo messages had insignificant words, such as citation, location and hyperlink, which are not in the user's words and may cause obstacles to sentiment analysis. These insignificant words were in specific patterns, hence we used regular expressions (Brzozowski, 1964) to filter them out.

3.2. Sentiment analysis

In order to determine the sentiment value (SV) for each Weibo message in the collected corpus, we used a lexicon-based sentiment analysis algorithm following Cao et al. (2014). The sentiment of a sentence is determined by attitudinal words, which express emotions or attitudes, in the sentence. In addition, the sentiment is modified by degree words and negation words. Hence, attitudinal word lexicon (AWL), degree word lexicon (DWL) and negation word lexicon (NWL) are essential for a lexicon-based method. In order to establish the AWL in this study, first we integrated two widely-used AWLs, including Hownet (Dong and Dong, 2003) and National Taiwan University Sentiment Dictionary (NTUSD) (Ku et al., 2006). Then, we learned from Jiang et al. (2016a) who provide a domain specific AWL for large hydro projects. The final AWL in this study contained 5774 positive attitudinal words and 10,196 negative attitudinal words. DWL and NWL were directly inherited from Hownet, which contained 182 degree words and 20 negation words. Each attitudinal word has a sentiment value (SV_a) representing the polarity of its sentiment, each degree word has a modification coefficient (MC_d) for amplifying or diminishing the sentiment value of an attitudinal word, and each negation word inverses the sentiment value of an attitudinal word.

The sentiment value of a Weibo message was calculated as follows. First, we divided the message into several clauses according to the punctuation. Then, we searched attitudinal words in each clause. Once we found an attitudinal word, we continued to search degree words and negation words before the attitudinal word (and after the last attitudinal word). Next, according to the relative positions of the three types of words, we built four rules to determine the modified sentiment value (SV_m) of the attitudinal word, as shown in Table 1 (Cao et al., 2014). After that, the sentiment value of a clause was determined by adding up modified sentiment values of attitudinal words in that clause. Finally, the sentiment value of the whole message was the weighted average of all clauses in the message, emphasizing the weights of the first and the last clauses in the message. If the sentiment value of a message was less than zero, the message was classified as a negative message. Otherwise, the message was classified as a non-negative message.

For validating the accuracy of the sentiment analysis method, we randomly selected 2000 messages from the whole corpus. Three students majoring in hydraulic engineering annotated these 2000 messages. The messages, to which all three annotators assigned the same tags (negative or non-negative), were integrated into a standard data set for evaluating the algorithm. The Fleiss's Kappa statistic (Conger, 1980) for the manual annotation result is 0.841 and the standard data set contained 1762 messages. The accuracy of sentiment classification algorithm is 87.9%, which is similar to the Fleiss's Kappa statistic, indicating that the algorithm is effective comparing with human annotators.

3.3. Topic modeling

The collected Weibo messages were classified into two categories, including a negative category and a non-negative category, by the

Table 1
Rules for sentiment value modification.

Pattern	Example	Rule
Degree word + attitudinal word	Very safe	$SV_m = SV_a * MC_d$
Negation word + attitudinal word	Not safe	$SV_m = -1 * SV_a$
Negation word + degree word + attitudinal word	Not very safe	$SV_m = -1/3 * SV_a * MC_d$
Degree word + negation word + attitudinal word	Very not safe	$SV_m = -1 * SV_a * MC_d$

sentiment analysis algorithm. Topic modeling was used to reveal major topics in the two categories respectively.

Topic modeling is a text mining approach that allows us to detect the latent thematic structure of a set of documents (Blei, 2012). In our analysis, Latent Dirichlet Allocation (LDA) (Blei et al., 2003), the simplest and most commonly used topic model, is employed. LDA describes the generative process of a corpus. The basic idea behind LDA is that documents blend multiple topics, each of which is a distribution over a fixed vocabulary (De Battisti et al., 2015). For example, an article related to environmental impact assessment may have two topics, including "environment" and "public participation". The "environment" topic may have words related to environment, such as "environment", "pollution" and "protection", with high probability, and the "public participation" topic may have words related to public participation, such as "public", "participation", "engagement" and "policy", with high probability. In LDA, documents are generated by the following steps. Consider a corpus C contains D documents. For the jth document in the corpus C, a multinomial distribution θ_j is sampled from a Dirichlet distribution α to represent latent topics in the document. Then, for each word in the document, a topic z is selected from θ_j and the word w is generated from a topic-specific multinomial distribution φ_z which is sampled from a Dirichlet distribution β. The task of topic modeling is the use of the observed corpus C to infer the latent variables, including the posterior topic distribution (θ_j) for each document and the posterior term distribution (φ_z) for each topic.

The inference algorithms fall into two categories, which use different methods to approximate the true posterior distributions. The first category considers a Markov Chain, which is a sequence of random variables only dependent on the previous, and the limiting distribution is the posterior distribution (Griffiths and Steyvers, 2004). The second category transforms the inference problem into an optimization problem and solves the problem by variational methods (Blei et al., 2003). In this study, we used variational expectation-maximization (VEM) algorithm provided by the topic model package (Hornik and Grün, 2011) in R to perform the inference task. In order to determine the number of topics, which is the only parameter in an LDA model, we ran several models with different numbers of topics and tried to select the model with the most semantics. Previous studies show that reliance on statistical measures may lead to the selection of less meaningful models (Levy and Franklin, 2014). Hence, we relied more on qualitative analysis, which was based on manual inspection of the most distinguishable terms and close reading of the representative articles of each topic, to attain our preferred models shown in this paper. We also ran multiple iterations of selected models starting from different seeds and found quite consistent modeling results.

3.4. Comprehensive indicators

Three types of indicators, including Post Intensity (PI), Sentiment Polarity (SP) and Topic Proportion (TP), are constructed to describe the profile of the online public opinion on the TGP.

PI describes the intensity of online discussion on the TGP and contains two indicators, including Absolute Post Volume (APV) and Normalized Post Volume (NPV). APV is the absolute number of messages which reflects the absolute intensity of discussion on the TGP in the social media site. However, when analyzing the spatial (provincial) distribution of the posting behavior, the number of Weibo messages related to the TGP in a province is influenced not only by the prominence of the TGP related topic in that province but also by factors such as population sizes, levels of internet penetration and overall Weibo popularity. In order to eliminate the effect of the latter factors and to establish a normalized indicator, namely, NPV, the number of the TGP related Weibo messages for a province was divided by the "background post intensity" (Kirilenko and Stepchenkova, 2014) for that province. The "background post intensities" in different provinces were inferred by relative post volumes of messages containing common keywords, such "eating", "working" or "television", among different provinces. These data can also be obtained by an official API of Weibo.

SP represents the public sentiment to the TGP and is also presented by two indicators, including Negative Proportion (NP) and Average Sentiment Value (ASV). The former is a practical indicator containing value judgment, indicating the proportion of negative messages. The latter is a pure value and is more suitable for comparison.

Spatial and temporal distributions of PI and SP were analyzed to present the heterogeneities of TGP related discussion across the 34 provinces (including municipalities, autonomous regions and special administrative regions) in China and during the 24 months (June 2013–May 2015).

TP indicates the proportion of each topic identified by topic modeling. This indicator contributes to the content analysis of the collected Weibo messages. As we built two topic models for the sets of negative messages and non-negative messages respectively, TP can reflect the major public concerns on the negative impacts triggered by the TGP and the major factors which make the public enjoy the benefits provided by the TGP.

4. Results

4.1. The conflictive Three Gorges Project

TGP is the world's largest hydropower project and is also one of the most high-profile infrastructure projects in China. The construction of the project started in 1994 and completed in 2009 with a total dynamic investment of 248.5 billion Chinese Yuan. After completion, the 181-meter concrete gravity dam raises the normal pool level of the Three Gorges Reservoir (TGR) to 175 m above the sea level, creating a 660-kilometer-long and one-kilometer-wide lake along the Yangtze River (Jiang et al., 2016b), as shown in Fig. 2. TGP provides plenty of benefits including flood control, power generation, navigation, irrigation, tourism and carbon emission reduction (Tang et al., 2013). These benefits make the TGP become a symbol of China's infrastructure construction achievement. However, the TGP also causes significant environmental and ecological impacts, including increasing geological risks of reservoir-induced earthquakes and landslides (Chen, 1999), water body pollution (Fu et al., 2010), disruption of the riparian ecosystem (Wu et al., 2003), unstable reshaping of the Yangtze River system (Yang et al., 2011) and local climate change (Deng et al., 2012), on the Yangtze Valley. In addition, construction of the TGP causes nationwide social impacts, including 1.2 million involuntary immigrants (Tullos, 2009), imposed tax for financing (Jiang et al., 2016a) and serious corruption (Jackson and Sleigh, 2000).

All these benefits and impacts have made the TGP a "spectacle" in both media and academia. In the China Core Newspapers Full-text Database which collects 605 national and local newspapers in China, we found over 8000 news articles with a title including "Three Gorges" since 2000; in the China Academic Journals Full-text Database which collects 10,080 academic journals in China, we found over 23,000 academic articles with a title including "Three Gorges" since 1945; and in the Web of Science, we found over 700 academic articles with a title including "Three Gorges" since 1994. The three search tasks were conducted in November 2015 and the three databases are all provided by Chinese National Knowledge Infrastructure (CNKI.net) (Cao et al., 2002).

In addition to these elite opinions, the rapid development of social media services in China provides platforms for common people discussing the TGP. The controversial nature makes the TGP a continuously hot topic in China's cyberspace (Jiang et al., 2016a). Assessment of the online public opinion of the Three Gorges Project is a valuable supplement to the ex post evaluation of the project, helps the project better exert its multiple functions, and provides suggestions for future infrastructure project management, especially public participation management. The assessment results are as follows.

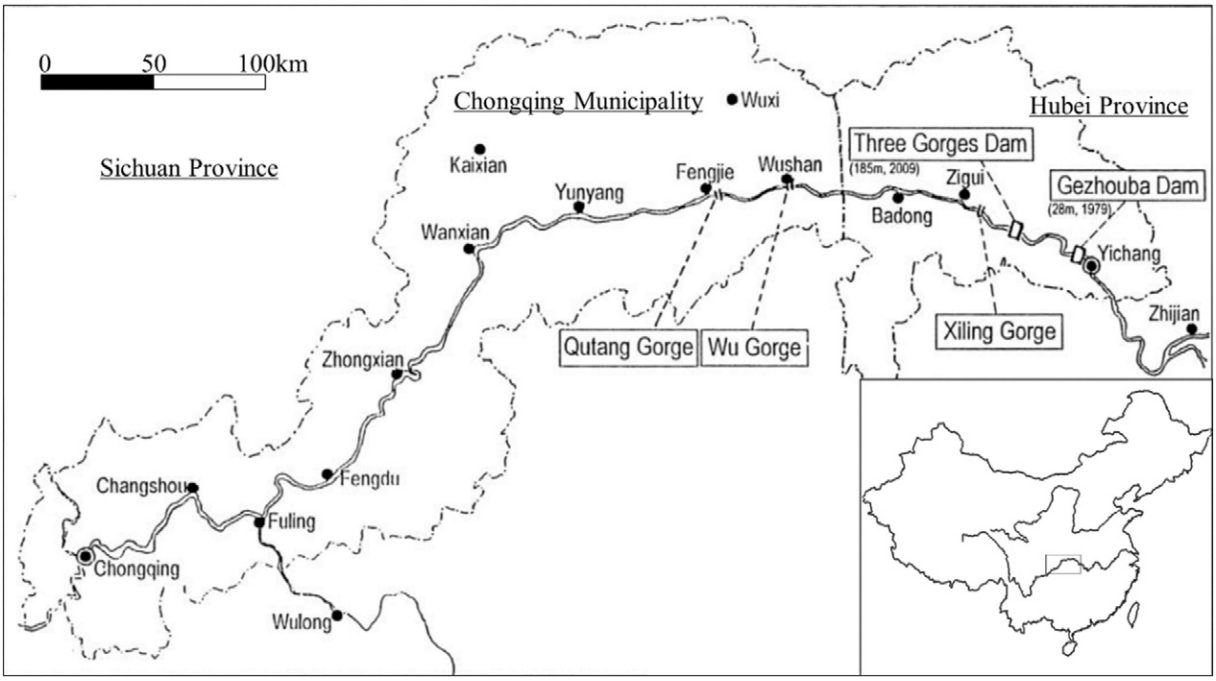


Fig. 2. The Three Gorges Reservoir area.

4.2. Assessment of post intensity (PI)

The post intensity of Weibo discussion on the TGP varied among the 34 provinces in China. Fig. 3(a) presents the APV of each province in China and Fig. 3(b) presents the NPV of each province. In Fig. 3(b), provinces with the average NPV have the value equaling to 1.

In terms of APV, Beijing (19,908), Hubei (15,747), Guangdong (15,302), Shanghai (11,638) and Jiangsu (9291) are the topic five provinces. Messages from these five provinces dominate the TGP related discussion on Weibo, accounting for 41.9% of all collected messages. Hubei is the province where the TGP is located. Jiangsu and Shanghai are the two most downstream provinces in the Yangtze Basin. Beijing, Shanghai and Guangdong are the three most prominent provinces in China in terms of economic, political and cultural strengths. Meanwhile, Macau (1060), Taiwan (916), Ningxia (695), Tibet (586) and Qinghai (581) have produced the least TGP related messages. Ningxia, Tibet and Qinghai are underdeveloped provinces in western China. Macau and Taiwan are relatively developed. However, Macau has a population of only 0.6 million. Taiwan has different political system and people in Taiwan use different social media services from the mainland China. In fact, these five provinces also have the lowest "background post intensities".

In terms of NPV, the top five provinces are Macau (3.92), Hubei (2.64), Tibet (2.05), Beijing (1.53) and Shanghai (1.40). It reflects that the TGP related topics are more dominant in the public discourse of the five provinces. Meanwhile, the five provinces with the lowest NPV are Liaoning (0.64), Jilin (0.59), Heilongjiang (0.56), Fujian (0.54) and Guangxi (0.51). These five provinces are far from the TGP both geographically and socio-economically.

Temporal analysis shows that the TGP is a persistently hot topic in Weibo. The monthly average number of Weibo messages about the TGP is 7151, including 3293 negative messages and 3858 non-negative messages.

As shown in Fig. 4, the discussion of the TGP on Weibo exhibits high temporal variability. The monthly trend of APV presents six peaks during the two years period. These peaks are partially explained by major online news events related to the TGP. Table 2 lists these news articles obtained from Baidu News (a Chinese online news aggregator). Some news articles present negative events about or skeptical comments on the TGP, such as the misused funds, privatization of the power generators, military security risks, construction corruption and earthquake impacts. Other news articles present positive events related to the TGP, such as flood control function, tourist discount and group wedding.

4.3. Assessment of sentiment polarity (SP)

For the whole collected Weibo messages, the overall NP is 0.460 and the overall ASV is −0.121. Figs. 5(a) and (b) present the NP and ASV of the TGP related Weibo messages originating from each province in China respectively. A higher NP proportion means a more negative emotion, expressed by the Weibo discussion in a province, to the TGP and vice versa. In contrast, a higher ASV refers to a less negative emotion.

Shandong (0.526), Shanghai (0.522), Beijing (0.519), Shanxi (0.504) and Tianjin (0.500) are the top five provinces with the highest NP. Macau (0.201), Tibet (0.307), Hubei (0.309), Qinghai (0.324) and Taiwan (0.336) have the five lowest NP.

In terms of ASV, Hubei (0.123), Macau (0.102), Taiwan (0.079), Qinghai (0.071) and Tibet (0.061) are the top five provinces with the highest ASV. Shandong (−0.224), Shanghai (−0.210), Liaoning (−0.201), Tianjin (−0.183) and Shanxi (−0.181) have the five lowest ASV. NP and ASV are highly correlated ($N = 34$, $r = -0.96$, $p < 0.001$).

The two indicators of SP also present high temporal variability with a slight seasonality. As shown in Fig. 6, both non-negative proportion (1-NP) and ASV reach to a trough in summer (July and August) and winter (December), and reach to a spike in autumn (September and October) and Spring (May).

4.4. Assessment of major topics

Two topic models are established to fit the corpora of negative messages and non-negative messages respectively. Both models contain 37 topics. Tables 3 and 4 show the first 20 topics (with a relatively higher TP) of the two models respectively. Terms shown in Tables 3 and 4 are the ten most frequent terms, which have been translated from Chinese to English, of each topic. It should be noted that the two search keywords, i.e. "Three Gorges Project" and "Three Gorges Dam" have been removed before topic modeling for more informative results. Topics in Tables 3 and 4 are sorted according to their overall TP, from the highest

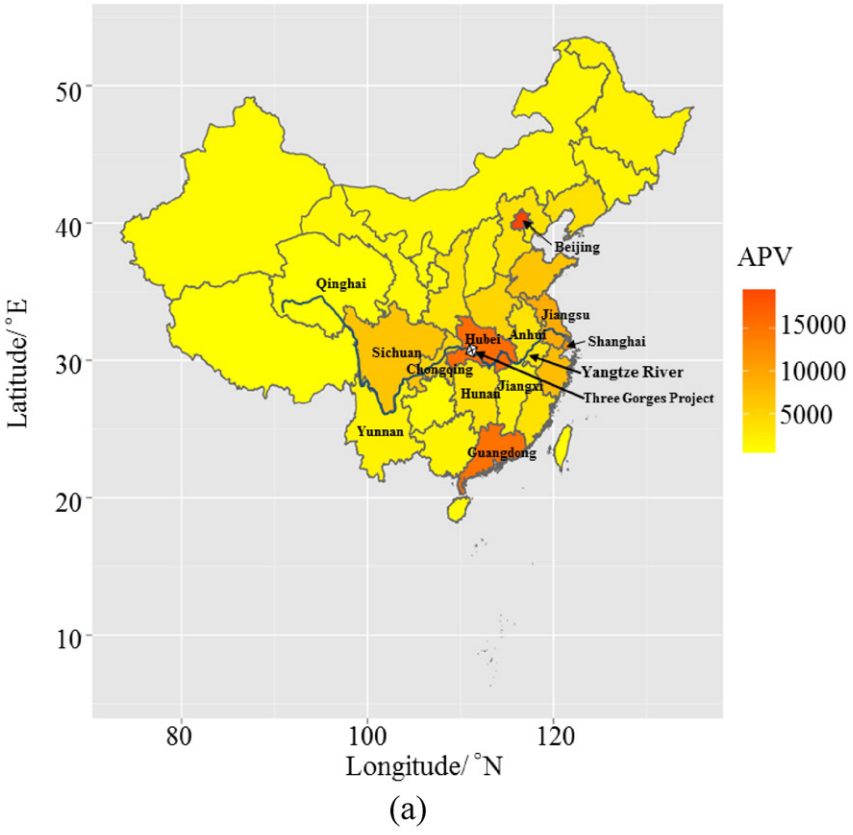


Fig. 3. Spatial distributions of post intensity in terms of (a) APV and (b) NPV.

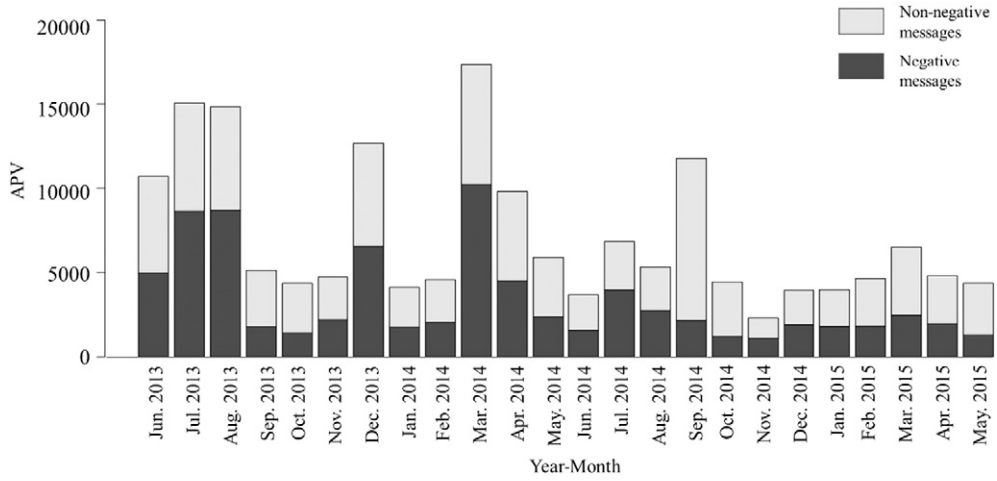


Fig. 4. Temporal trends of APV (Jun. 2013–May 2015).

to the lowest. Based on our prior knowledge on the TGP, these topics are readily recognizable and related to major issues about the TGP. We provide interpretations of some typical topics as follows:

4.4.1. Topics in the negative corpus

Topic N_1 is the largest topic accounting for 4.60% of the negative corpus. Topic N_1 contains words like "destroy", "ecology", "environment", "influence" and "disaster", and thus clearly pertains to TGP's negative impacts to the environment and ecology. Note that the "South-to-North Water Transfer Project (SNWTP)", which is the largest water diversion project in China, also appears in this topic. It indicates that Weibo users often compare the TGP with the SNWTP, when discussing the environmental and ecological impacts of the TGP.

Topic N_2 is the second largest topic accounting for 4.26% of the negative corpus. It contains many geographic names like "Three Gorges", "Yichang City", "Chongqing City", "Yangtze River" and "Poyang Lake". These geographic names are highly related to the Yangtze River and the TGP. Yichang City is the site where the TGP is located and Chongqing City covers the major part of the 660-kilometer TGR. Poyang Lake, the largest freshwater lake in China, connects to the Yangtze River and is downstream from the TGP. Considering that "water level" is also a high-frequency term in this topic, this topic mainly addresses the impacts of the TGP on the Yangtze River system.

Topic N_3 and Topic N_10 discuss the geological risks associated with the TGP and account for 3.97% and 3.01% of the negative corpus respectively. Topic N_3 contains words like "earthquake", "happen", "Zigui County", "impoundment" and "correlation" and apparently discusses the correlation between the impoundment of the TGP and the twice earthquakes happened in Zigui County, which is just 30 km away from the TGP, in 2014. Topic N_10 contains words like "hidden peril", "fracture", "Qiyue Mount", "emerge" and "man-made calamity" and thus addresses the correlation between the TGP and the fracture in Qiyue Mount which is near the TGR area.

Wanli Huang, who was a famous hydraulic engineering professor in Tsinghua University (the top engineering and technology university in China) and opposed to the construction of the TGP, appears in Topic N_4, Topic N_7 and Topic N_20. The three topics account for 9.59% of the negative corpus. Topic N_4 mainly focuses on Professor Huang's predictions of the TGP. Topic N_7 and Topic N_10 mainly discuss Professor Huang's opposing activities to the construction of the TGP.

In addition to Wanli Huang, Zhirong Liu and Luowei Wang, as two major critics of the TGP currently, appear in Topic N_8 and Topic N_11 respectively. Mr. Liu criticized the TGP from the perspective of national economy assessment. He thought that the financial process of the TGP was not reasonable and the payback period would be longer than expected. Mr. Wang criticized the TGP from the perspective of climate change. He thought that the construction of the TGP triggered the climatic anomaly in the middle and lower reaches of the Yangtze Basin. Their critical articles are mainly published on Sina Blog, which provides a convenient interface for reprinting blogs to Weibo.

Other topics are also readily recognizable and are related to negative impacts of or critical comments on the TGP. Topic N_6 focuses on the embezzlement of the funds for involuntary immigrants of the TGP. Topic N_17 discusses TGP's negative impacts, including seawater intrusion and sediment reduction, on Shanghai. Topic N_18 points out the security risks of the TGP in potential military actions.

4.4.2. Topics in the non-negative corpus

Contents of topics in the non-negative corpus are more concentrated than that in the negative corpus. Tourism related topics, such as Topics P_1, P_2, P_4, P_5, P_7, P_12, P_15, P_17, P_19, P_20 and others not shown in Table 4, account for nearly 40% of the whole non-negative corpus. Some of these topics discuss detailed scenic spots, such as "Tribe of the Three Gorge" in Topic P_2, "Dragon Spine Rock" in Topic P_4, "Wujia Hill" in Topic P_17 and "Xinling Gorge" in Topics P_5 and P_20. Topics P_1, P_15 and P_19, with words like "feel", "happy", "route", "visit" and "activity" address tourism activities and feelings. The free admission

Table 2
Major news events in the six peak months.

Date	Major news event
2013-6-7	Auditing reports revealed misused funds in the TGP.
2013-6-22	Guangdou Zhang, one of the most important hydropower specialists in China and involved in the design of the TGP, died.
2013-7-11	Sharp increase of the inflow discharge came to the TGR, with a peak of 41,000 m^3/s.
2013-7-16	Seven tourists got thunderstruck in the scenic area of the TGD.
2013-7-23	Have the power generators in the TGP been privatized?
2013-8-1	The TGD offered a tourist discount to Yichang citizens.
2013-8-28	Vice Premier Gaoli Zhang chaired the plenary session of the TGP Construction Committee.
2013-8-28	A hundred of couples held the wedding together on the TGD.
2013-12-2	The TGD may have military security risks.
2013-12-16	Eight conflictive issues of the TGP
2013-12-16	Does the TGP trigger earthquakes?
2013-12-22	The TGP has recovered the cost.
2014-3-01	Construction corruption of the TGP has been revealed.
2014-3-30	The 4.7-magnitude earthquake in Zigui, Hubei did not impact the TGD
2014-9-16	The TGP started a new-round full-capacity storage test with water level of 175 m.
2014-9-25	The TGD offered free admission to all ethnic Chinese tourists, including those from overseas.

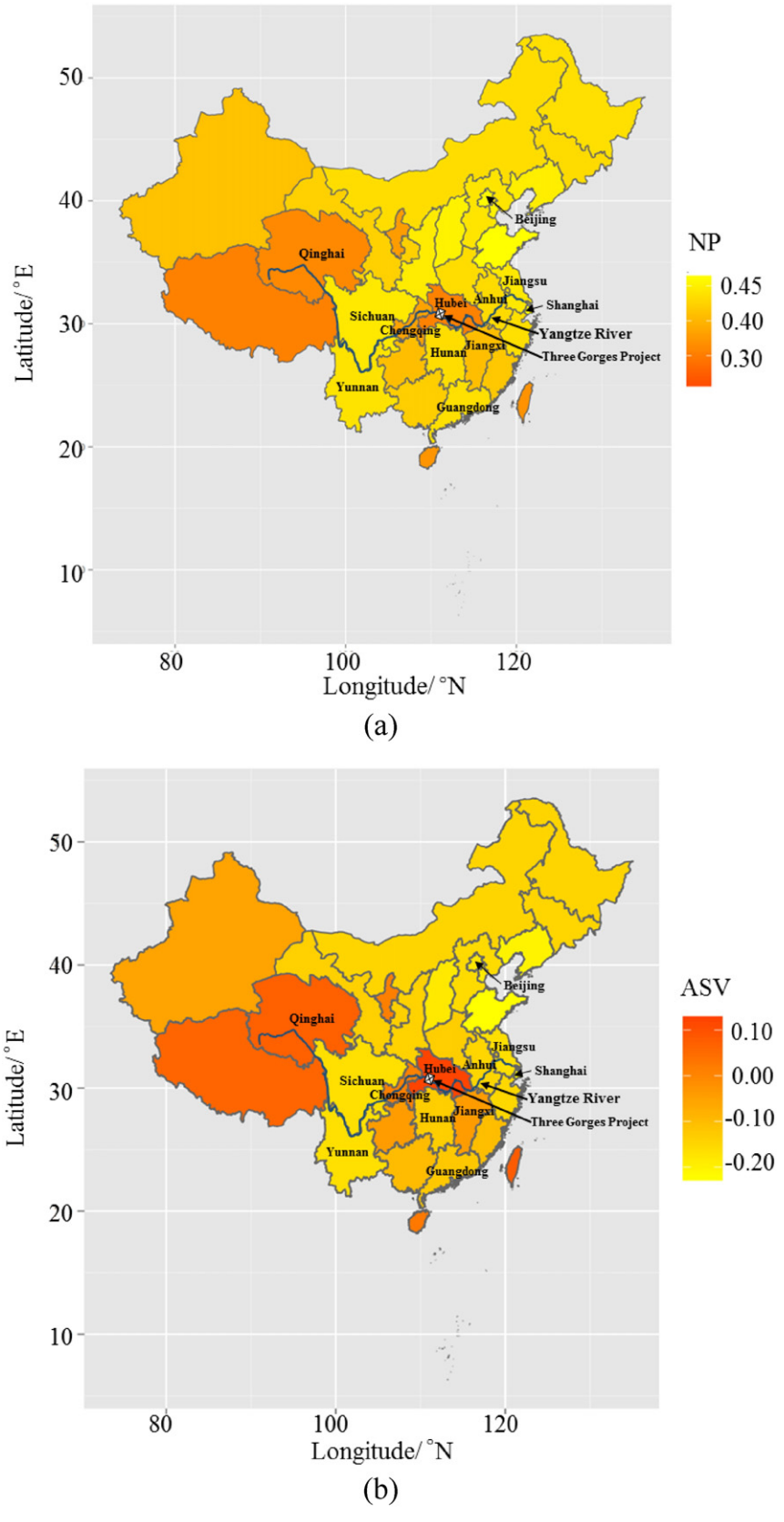


Fig. 5. Spatial distributions of sentiment polarity in terms of (a) NP and (b) ASV.

offered by the TGD to all ethnic Chinese tourists (also indicated in Table 2), are revealed in Topic P_7 and Topic P_12.

In addition to tourism, some topics focus on praising the great achievement of the TGP. For instance, Topic P_3, accounting for 4.95% of the non-negative corpus, contains words like “China”, “science”, “engineering”, “decision”, “expert” and “world”, and thus addresses the validity and rationality of the decision of the TGP. Topic P_11, accounting for 3.13% of the non-negative corpus, contains word like “great”,

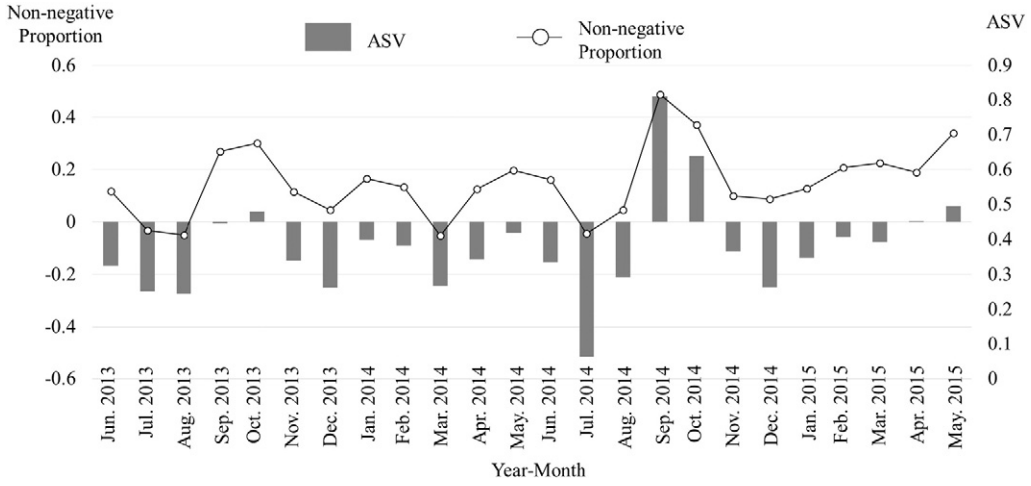


Fig. 6. Temporal trends of SP on the TGP (Jun. 2013–May 2015).

"human", "realization", "socialism", "engineering", "history", "centuries" and "present" and thus addresses the greatness and historical significance of the TGP. Topic P_18, accounting for 2.43% of the non-negative corpus and with words like "ship", "ship lock" and "navigation", focuses on the navigation functions provided by the TGP. Topic P_14, accounting for 2.70% of the non-negative corpus, discusses the water storage capacity of the TGR and compares it with other reservoir or lakes, such as the reservoir of Hoover Dam and the Qinghai Lake.

Some topics in the non-negative corpus also address problems or impacts of the TGP, but with a more neutral tone. For instance, Topic P_3 contains "ecology"; Topic P_8 contains "problem", "immigrants"

Table 3
Major topics in the corpus of negative messages.

Number	Terms	TP
N_1	Destroy, ecology, engineering, SNWTP[a], environment, nature, influence, disaster, bring, human	4.60%
N_2	Three Gorges, Yichang City, Yangtze River, Poyang Lake, water level, Chongqing City, dam, see, feel, present	4.26%
N_3	Earthquake, Three Gorges, happen, Hubei, Zigui County, expert, impoundment, twice, kilometer, correlation	3.97%
N_4	Prediction, realization, Wanli Huang, problem, blow up, deteriorate, be forced, Mr. Huang, Yangtze River, ecology	3.62%
N_5	Sichuan, sneer, shocked, Chongqing City, built, climate, continuous, torrential rain, teacher, emergency	3.54%
N_6	Immigrants, money, billion CNY[b], embezzlement, final accounts, audit office, audit, engineering completion, usage, expenditure	3.37%
N_7	That year, Wanli Huang, commence, ship lock, capacity, oppose, hear about, final, famous, old gentleman	3.33%
N_8	Rongzhi Liu, Shanghai, compatriots, electricity, Three Gorges, from, common people, Academician, victim, statement	3.25%
N_9	Secret, China, exposure, hidden, influence, Japan, dam, terror, thorough, nature	3.08%
N_10	Hidden peril, fracture, Qiyue Mount, emerge, cost, know, appear, man-made calamity, disaster, believe	3.01%
N_11	Three Gorges, demolition, expert, laugh, water conservancy, dam, suggestion, demonstrate, Weiluo Wang, blow up	3.00%
N_12	Game, destroy, China, exist, unable, serious, image, conquer, task, amusing	2.85%
N_13	A kilowatt-hour, electricity price, that year, commence, promise, remember, government, should, fulfil, memory	2.83%
N_14	Think, Yangtze River, news, Three Gorges, Nanjing City, focus, terrible, Zhouqu County, solve, problem	2.82%
N_15	Three Gorges, people, whole nation, funding, interest group, machine, billion CNY, profit-making, transfer to, common people	2.81%
N_16	Problem, see through, focus, impoundment, eight main, geology, disaster, earthquake, trigger, reservoir area	2.77%
N_17	Shanghai, victim, CAS[c], expert, influence, Yangtze River, flow backward, seawater, sediment, pollution	2.76%
N_18	China, United States, military, bomb, consideration, Three Gorges, Taiwan, mainland, amazing, secret	2.75%
N_19	Check, Gezhou dam, extinction, today, flood releasing, China, time, video, first, designer	2.66%
N_20	Oppose, Three Gorges, unsolved, Wanli Huang, water conservancy, SNWTP, report, expert, that year, construction	2.64%

[a] South-to-North Water Transfer Project.
[b] China Yuan.
[c] Chinese Academy of Science.

Table 4
Major topics in the corpus of non-negative messages.

Number	Terms	TP
P_1	Yichang City, check, smile, today, see, feel, present, happy, know	6.41%
P_2	Yichang City, Three Gorges, Chongqing City, Tribe of the Three Gorges, tour, Yangtze River, Qu Yuan, cruise ship, Hubei, Zigui County	5.09%
P_3	China, Three Gorges, Science, engineering, ecology, Yangtze River, decision, experts, SNWTP[a], world	4.95%
P_4	Yangtze River, city, Yichang City, Dragon Spine Rock, ait, west, action, location, hurry, Hubei Province	4.84%
P_5	Xiling Gorge, Gezhou Dam, Three Gorges, kilometer, distance, beauty, Yichang City, located in, scenic area, starting point	4.03%
P_6	Recalcitrant, that year, bluntly, simple, two steps, world, power generation, hydropower plant, think, built	3.97%
P_7	Ticket, China, foreigner, tourist, scenic area, free, free ticket, scenic spot, charging, overseas	3.96%
P_8	China, problem, Three Gorges, immigrants, Poyang Lake, engineering, group, government, responsible for, economy	3.47%
P_9	Expert, Three Gorges, China, influence, climate, tell, Shanghai, journalist, happen, Yangtze River	3.46%
P_10	Ha-ha, China, engineering, hydropower, commence, ending, know, railway, Rui Li, power	3.18%
P_11	Great, earthquake, human, realization, socialism, engineering, history, centuries, Sichuan, present	3.13%
P_12	Tourists, tour, scenic area, free, scenic point, ticket, Tanzi Mount, China, open, commemorate	2.73%
P_13	Focus, compatriot, wonderful, free ticket, Wechat, leisure, new style, river closure, gentleman, buy	2.72%
P_14	Water-storage capacity, billion cubic meters, earth, United States, dam, see, flood, Hoover, big lake, Qinghai Lake	2.70%
P_15	Three Gorges, Yichang City, old people, route, tour, Hunan, Hubei, welcome, grand, Wuhan City	2.61%
P_16	Engineering, main body, world, power generation, located in, territory, Hubei Province, part, total investment, Yichang City	2.57%
P_17	Impoundment, water level, Yangtze River, toll station, Wujia hill, Three Gorges, currently, enter, navigation channel, upstream	2.43%
P_18	Ship lock, Yangtze River, economy, Yichang City, Three Gorges, navigation, Chongqing City, five-level, ship, engineering	2.43%
P_19	Visit, minority, Taiwan, first, activity, flood releasing, visit learning, video, wedding, morning	2.41%
P_20	Xiling Gorge, Yangtze River, dam, Hubei Province, Zigui County, built, famous, located in, a piece of, Yichang City	2.38%

[a] South-to-North Water Transfer Project.

and “Poyang Lake”; and Topic P_11 contains “earthquake”. But no word with a significant negative emotion is found in these topics.

Ironic tone can be found in some topics. For example, a representative message of Topic P_6 reads “Bluntly, it is very simple. They used our money to build the TGD and then use the TGD to earn money from us. If you are recalcitrant, they will take you away!” These ironic messages actually present negative emotions to the TGP. However, the lexicon-based sentiment analysis method is weak in recognizing the ironic tone. It is also one of the major causes for sentiment classification errors.

We also find a topic with advertising information. Topic P_13 contains words like “focus”, “compatriot”, “wonderful”, “free ticket”, “Wechat”, “leisure”, “new style”, “river closure”, “gentleman” and “buy”. It means that some online merchants inserted the TGP related news events into their advertising messages to heighten the visibility.

As a summary, Tables 3 and 4 display the key terms and TP values of major topics in the negative corpus and the non-positive corpus respectively. With these results, decision-makers can easily know the factors which make people dissatisfied or satisfied with TGP and the importance of these factors.

5. Discussion

5.1. Implications for the TGP

Three Gorges Project is one of the most controversial infrastructure projects of significance to China in many aspects, including politics, economy, national safety and environment. Therefore, it is important to assess the public opinion about the TGP.

The search result demonstrates that the TGP is a constantly attractive topic on China's online public sphere. Weibo users are keen on expressing their opinions about the TGP and averagely produce over 200 messages per day. It further validates that the TGP is a “spectacle” not only in academia and traditional media but also in the newly-emerging social media.

The spatial pattern of PI reveals that Weibo users from Shanghai, Beijing and Hubei are more interested in the discussion of the TGP. The superiority in the three provinces is reflected in both APV and NPV and is seemingly reasonable. Hubei is the place where the TGP is located. Shanghai, located at the estuary of the Yangtze River, is the largest city in China and is deemed to be as a “victim” of the TGP (see Topic N_17). Beijing is geographically distant from the TGP and the Yangtze Basin, but it is China's political, cultural and educational center usually producing a large number of critical comments on social issues. The high values of NPV in Macau and Tibet can be attributed to their minimal “background post intensities”, which may make the results highly sensitive and possibly unreliable.

Analysis of SP indicators shows that slightly less than half (46.0%) of Weibo messages related to the TGP show negative sentiment in total. This is a relatively cheerful result for the managers of the Three Gorges Corporation, as previous studies find that internet users tend to report more negative comments, especially those related to public issues (Suler, 2004; Lapidot-Lefler and Barak, 2012).

The spatial pattern of SP reveals two paradoxes. Hubei and Chongqing seems to be the two most negatively affected provinces by the TGP, as the TGD is located in Hubei and the TGR submerges certain areas of Hubei and Chongqing. In other words, most of the identified or rumoured negative impacts, such as involuntary immigrants, water body pollution, landslides and reservoir-induced earthquakes, of the TGP will particularly harm the interests of the two provinces. However, sentiment analysis shows that both Hubei and Chongqing have a relatively low NP (third and eighth lowest respectively) and a relatively high ASV (first and eighth highest respectively). In fact, if we exclude those provinces with small APV (fewer than 1200 messages), Hubei and Chongqing will be the two provinces with the most positive sentiment to the TGP. A possible explanation includes the following three reasons. First, the TGR's flood storage capacity is so vital that can reduce the frequency of major downstream flooding from once every 10 years to once every 100 years (Xie et al., 2013). Many riparian cities in Hubei, including the capital Wuhan, greatly benefit from the flood control function of the TGP. Second, the construction of the TGP and the operation of the power plant create many job opportunities for the two provinces. Third, people in Hubei and Chongqing are more convenient and likely to travel to the scenic spots around the TGP than people in other provinces. Positive emotions are more likely to be expressed by these visitors from Hubei and Chongqing. The dominance of tourism related topics in the non-negative corpus also partially supports this explanation.

The other paradox involves the two most downstream provinces, i.e. Jiangsu and Shanghai, in the Yangtze Basin. As is well known that flood control and power generation are the two most significant benefits provided by the TGP (Zheng et al., in press). Shanghai and Jiangsu enjoy the two benefits and are hardly affected by the aforementioned negative impacts caused by the TGP (excluding the reduction of sediment downstream and superflood caused by TGD's collapse). However, both Shanghai and Jiangsu have a relatively high NP (second and ninth highest respectively) and a relatively low ASV (second and ninth lowest respectively). We provide an explanation as follows. The prominence of negative emotion in Shanghai seems to be induced by the heavily reprinted comment, which states that Shanghai is the biggest “victim” of the TGP due to sediment reduction (seen in Topic N_17). Many cities in southern Jiangsu are riparian cities which may also suffer from the sediment reduction. In addition, these cities are adjacent to Shanghai and have a shared cultural identity with Shanghai. Hence, people from these cities are easy to empathize with such negative comments.

The temporal trend of PI presents that major news events affect microblogging behavior on the TGP in China. This is in accordance with Kirilenko and Stepchenkova's study (Kirilenko and Stepchenkova, 2014), which find a similar effect of news events on tweeting on climate change. Based on topic modeling results, we can find corresponding topics in Tables 3 and 4 to most news events shown in Table 2. The slight seasonality in temporal trend of SP may be explained by comprehensive reasons. First, May, September and October are the tourist seasons for the TGD and scenic spots around it. Therefore, more positive emotions are expressed in these months. Second, July and August are in the main flood season of the middle reaches of the Yangtze (Li et al., 2015). Misunderstanding of TGP's flood control function often emerges in these two months and triggers negative comments on the TGP. Finally, December is the month for annual summary and audit in China. Hence, criticism of the TGP is likely to appear.

Topic modeling of the negative and non-negative messages uncovers Weibo users' major concerns of the TGP. Most of the TGP's environmental impacts discussed by Weibo users are in accordance with the major academic concerns of the TGP (Jiang et al., 2016b). However, in terms of social impacts, academic articles mainly focus on land use changes and immigrants in the Three Gorges Area. Meanwhile, Weibo users are concerned with broader issues, such as engineering corruption, military security, rationality of the decision process, and anecdotes of related scientist (Professor Huang Wanli). The dominance of tourism related topics in the non-negative corpus reveals that tourism is important to the TGP in terms of improving public reputation. Comparing with the three highly identified functions, namely, flood control, power generation and navigation improvement, tourism is a marginal function of the TGP. However, it earns most of the favor in social media for the TGP.

5.2. Implications for large infrastructure project post evaluation

In a democratic society, the public opinion is very critical to policy making in relation to public projects (Ji and Hong, 2016). The proposed method is valuable for government officials, company managers and researchers better assessing the public opinion on a large infrastructure

project in the ex post evaluation stage. Based on the developed indicators, the public satisfaction, which is an important subjective factor in ex post evaluation framework, can be comprehensively measured, providing an opportunity for decision-makers to establish more democratic ex post evaluation framework for large infrastructure projects. Some basic managerial recommendations are provided as follows:

(1) Encourage the general public to participate in the online discussion of large infrastructure projects. Social media platforms provide an interface for the general public expressing their feelings, attitudes and appeals towards a project to exercise their "Right to know", "Right to participate", and "Right to supervise", as ensured and guaranteed by law (Li et al., 2016). This interface is also an aggregator for government officials and project managers collecting the public opinion on controversial events with much lower costs and higher flexibilities than traditional public participation approaches, such as demonstration meetings, public hearings, seminars and other forums (Bertot et al., 2012). However, the representativeness of online public opinion must be ensured by a large number of online participants. Hence, it is important to encourage more people to participate in public issues by online approaches.

(2) Monitor of the online presence of the project constantly (He et al., 2013). As the improvement of people's livelihood is becoming more important in recent project evaluations (Nakamura, 2000), the significance of public satisfaction in ex post evaluation framework should be correspondingly raised. Large infrastructure projects often have long-term impacts on the environment and society, which ultimately affect the livelihood of the public and stimulate the public to discuss the project in the public sphere, particularly via social media (Bertot et al., 2012; Bonsón et al., 2012; Lee and Kwak, 2012). Hence, monitoring and analysis of social media data is an efficient way for decision-makers to get the information of the public satisfaction with the concerned project. In addition, as the public satisfaction with a large infrastructure project varies with time and space, using social media data seems to be the only economically feasible method to measure the spatio-temporal variations of the public satisfaction (Schober et al., 2016). Based on the proposed framework in this study, decision-makers can understand the public satisfaction with the concerned project in a more comprehensive way. The assessment of public satisfaction is also an important reference for determining the relative weights of objective factors in ex post evaluations (Ji and Hong, 2016) and subsequently increases the significance of ex post evaluation practices to the general public.

(3) Establish benchmarks and topic systems. Benchmarks make post intensity and sentiment analysis results more meaningful in practice (He et al., 2013; Jiang et al., 2016a). Otherwise values of the indicators can only provide isolated values. Based on effective and realistic benchmarks, automatic warning and controlling mechanisms can be established with real-time monitoring data. Additionally, with abundant monitoring data, topic systems should be established for different kinds of infrastructure projects. These topic systems contain refined knowledge about the public opinion towards dams, railways, nuclear plants and other kinds of large infrastructure projects and will be easily reused by government officials and project managers in future practices to improve the efficiency of analyzing online public opinions on similar projects.

5.3. Limitations and future work

Any new form of management research or practice faces complications in execution. In our study, the limitations are shown in the following three aspects. First, the data were collected from only one source. Although Weibo.com is the most dominant microblogging site in China, various data sources will lead to a more reliable result. Second, the accuracy of the sentiment analysis method has the space to improve. The lexicon-based method employed in our study has a poor performance in processing messages with an ironic tone. Therefore a more in-depth understanding of the rules in the algorithm is required. Third, the topic modeling processes in our study is nontrivial to some extent, particularly in determining the number of topics. In previous studies, statistical measures, such as perplexity and entropy, have been proposed to trivially fit the model (Blei and Lafferty, 2007). But as these statistical measures were found to be less semantically meaningful (Grimmer and Stewart, 2013). In order to reveal interesting information as much as possible, we employed manual inspection to fit the model and obtained meaningful results. The contradiction of operability and interpretability are likely to be solved by a more structured model in future. In addition to overcoming these technical limitations, future work will mainly focus on investigating more cases to find general patterns of online public opinions on large infrastructure projects.

6. Conclusions

Large infrastructure projects are facing more and more challenges from their environmental and social impacts. These challenges could be tougher with a poor performance on public participation management and the public satisfaction with these impacts should be comprehensively assessed in the ex post evaluation stage. Social media's capability to facilitate interpersonal and group interaction highlights new and unique opportunities for government officials, company managers and researchers to inform or be informed by the citizenry (Golbeck et al., 2010). Policies and measures related to large infrastructure projects can be rapidly spread to the general public by social media. Meanwhile the public can also publish comments on the projects by social media to show their own opinions, which can be collected and analyzed by text mining technologies, and be understood by government officials, company managers and researcher from multiple dimensions. Subsequently, the public satisfaction, which is an important subjective factor in ex post evaluation framework of large infrastructure projects, can also be comprehensively described. This study proposes an assessment framework, which integrates sentiment analysis and topic modeling, to transform unstructured social media comments on large infrastructure projects into sentimental and topical indicators for enhancing practices of ex post evaluation and public participation. Through an in-depth case study which analyzes the online public opinion on the Three Gorges Project, this study demonstrates the effectiveness of the proposed assessment framework and indicators, and draws informative and valuable implications for understanding how Chinese people think about the world's largest hydropower project, which is of significance to ex post evaluation of the project.

Previous studies focus more on integrating experts' opinions in decision making of large infrastructure projects (Singh and Tiong, 2005; Shen et al., 2011). However, a scientific and democratic decision-making model should consider opinions from both experts and the public (Garland, 1999; Hampton, 2009). The proposed assessment framework is a response to the lack of effective methods that widely collect and comprehensively analyze the public opinion on large infrastructure projects in the literature. Only the interlinkage of experts' knowledge and public opinions provides a more comprehensive understanding of the complex and dynamic impacts caused by large infrastructure projects and lead to more effective practices (Schindler et al., 2016). Decision-makers can follow the proposed framework to acquire in-depth knowledge of the public opinion on a similar large infrastructure project. The proposed method is expected to be widely applied as a methodological strategy to assess public opinion in the ex post evaluation stage of large infrastructure projects in future.

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